Daily Combinatorics Drill #1

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Multiplication Principle · Factorials & Factorial Algebra · Permutations of Distinct Objects · Counting by Cases.

Section A Rapid Recognition ($\leq$15 s each)

Evaluate or count instantly. One rule per question. Pure recognition.

1. Evaluate: $5!$.

Ans: 120

2. Simplify: $\dfrac{8!}{6!}$.

Ans: 56

3. A menu offers 4 starters, 5 mains and 3 desserts. How many different three-course meals are possible?

Ans: 60

4. In how many ways can 4 different books be arranged on a shelf?

Ans: 24

5. Evaluate: $\dfrac{6!}{4!\,2!}$.

Ans: 15

6. How many 3-letter strings can be made from the 26 letters of the alphabet if letters may repeat?

Ans: $26^3 = 17\,576$

7. How many 3-letter strings can be made if no letter may repeat?

Ans: $26 \cdot 25 \cdot 24 = 15\,600$

8. Solve for n: $n! = 720$.

Ans: $n = 6$

9. How many two-digit numbers have both digits odd?

Ans: 25

10. Simplify: $\dfrac{(n+1)!}{(n-1)!}$.

Ans: $n(n+1)$

Section B Manipulation Drills ($\leq$50 s each)

Factorial algebra and direct counting. Set up the product, don't list outcomes.

1. In how many ways can 6 different people stand in a line? In how many of these arrangements does Amara stand first?

Ans: 720; 120

2. Simplify $\dfrac{n!}{(n-2)!}$ and hence solve $\dfrac{n!}{(n-2)!} = 42$.

Ans: $n(n-1)$; $n = 7$

3. A 4-digit code uses the digits 0–9. How many codes have no repeated digit?

Ans: $10 \cdot 9 \cdot 8 \cdot 7 = 5\,040$

4. How many three-digit numbers have all three digits different?

Ans: $9 \cdot 9 \cdot 8 = 648$

5. Gold, silver and bronze medals are awarded in a race with 8 runners. In how many ways can the medals be won?

Ans: $8 \cdot 7 \cdot 6 = 336$

6. How many arrangements are there of the letters of the word SPEED?

Ans: $\frac{5!}{2!} = 60$

7. A flag has 3 horizontal stripes. Each stripe is one of 5 colours, and adjacent stripes must be different colours. How many flags are possible?

Ans: $5 \cdot 4 \cdot 4 = 80$

8. A set has 6 elements. By deciding, for each element, whether it is in or out, count the subsets of the set.

Ans: $2^6 = 64$

9. Solve for n: $(n+1)! = 12\,(n-1)!$.

Ans: $n = 3$

10. A number plate consists of 2 different letters followed by 3 digits (digits may repeat). How many plates are possible?

Ans: $26 \cdot 25 \cdot 10^3 = 650\,000$

Section C Substitution & Structure ($\leq$90 s each)

Split into cases before you multiply. Watch for leading zeros and forced choices.

1. How many even three-digit numbers have all three digits different?

Ans: 328

2. Six people stand in a line. Two of them, Priya and Quinn, insist on standing next to each other. How many arrangements are possible?

Ans: $2 \cdot 5! = 240$

3. With the same six people, how many arrangements have Priya and Quinn *not* next to each other?

Ans: $720 - 240 = 480$

4. How many five-digit palindromes are there?

Ans: 900

5. Three married couples sit in a row of 6 seats so that each couple sits together. In how many ways can they be seated?

Ans: $3! \cdot 2^3 = 48$

6. Using the factorisation $360 = 2^3 \cdot 3^2 \cdot 5$, count the positive factors of 360.

Ans: $(3+1)(2+1)(1+1) = 24$

7. How many four-digit odd numbers have all four digits different?

Ans: $5 \cdot 8 \cdot 8 \cdot 7 = 2\,240$

8. $n! = 20\,(n-2)!$. Find n.

Ans: $n = 5$

Section D Challenge (SMC difficulty)

Insight over computation. If the last ones resist, sit with them — learn the tool, then return.

1. How many three-digit numbers consist of three different even digits and are divisible by 3? *(after SMC 2025 Q12)*

Ans: 20

2. How many rectangles (of any size, including squares) appear in a 3×3 grid of unit squares? *(after SMC 2022 Q3)*

Ans: 36

3. In how many ways can 8 people be split into 4 unlabelled pairs?

Ans: 105

4. In how many zeros does the number 100! end?

Ans: 24

5. Thirty squares are drawn side by side in a line. The first and last squares are shaded, and more squares must be shaded so that both rules hold: (a) no two adjacent squares are shaded, and (b) there are never more than three consecutive unshaded squares. What is the difference between the largest and smallest possible number of shaded squares? *(after SMC 2025 Q9)*

Ans: $15 - 9 = 6$